

Exploring User Preferences of VR Relaxation Experiences: A Comparative Mixed-Methods Study

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ABSTRACT

This comparative study presents the results of a user experience investigation conducted during the use of four relaxing VR applications, distinguished by style and interactivity (Fujii, Real VR Fishing, TRIPP, Nature Treks VR), tested by 12 participants in a balanced order, within a within-subjects scenario. Both immersion and anxiety were measured. Semi-structured individual interviews were conducted to better understand user preferences. Among the most frequently suggested features, personalization and voluntary microinteractions appeared to be the most preferred.

Index Terms: Virtual Reality—User Experience—Usability Testing—Relaxation; Anxiety—Immersion—Mixed-methods Study

1 INTRODUCTION

Virtual Reality (VR) can be an effective tool for the purpose of relaxation [1] and mindfulness [7], at the same time offering a safe and controlled environment where individuals can practice their speeches and learn to manage their anxiety under various speaking conditions [6]. The motivation for this study was to understand user preferences in the relaxing VR experiences as a step towards designing a relaxation mode for a new virtual reality application that combines training of public speaking in front of the virtual audience and training of voice emission in the relaxation voice exercises. The project was introduced in [3], and the recent progress was reported in [4].

2 MATERIAL AND METHODS

In order to choose the experiences to investigate, first, we reviewed 15 relaxation VR applications available on the market (Nature Treks VR, Real VR Fishing, Zen VR, Musical Relaxation- Tetris Effect: Connected, TRIPP, Fujii, Nature Relaxation theBlu, Nature Treks: Together, Luna, Hand Particle, Flowborne VR - Biofeedback Breathing Meditation, Squingle, Visitations VR, Morph VR, Sheaf - Together EP). Four of them was selected, using criteria of providing various graphics styles and interactivity levels. **Fujii** experience combines elements of outdoor adventure with creative gardening. **Real VR Fishing** is a very realistic VR fishing experience, which allows the player visit various places and fish there. **TRIPP VR** is a meditative experience that allows one to focus and calm down through meditations, simple games, and fractal visualizations. In **Nature Treks VR**, the player wanders through nature scenes such as, a forest, beach, meadow, savannah, or space and has the ability to edit/customize the place by, for example, planting trees or flowers.

In order to measure immersion in each experience, a Polish adaptation [10] of the **Immersion Questionnaire** [5] was used. It was modified by removing two questions: "How much did you want to win the game?" and "Did you feel uncertain about whether you

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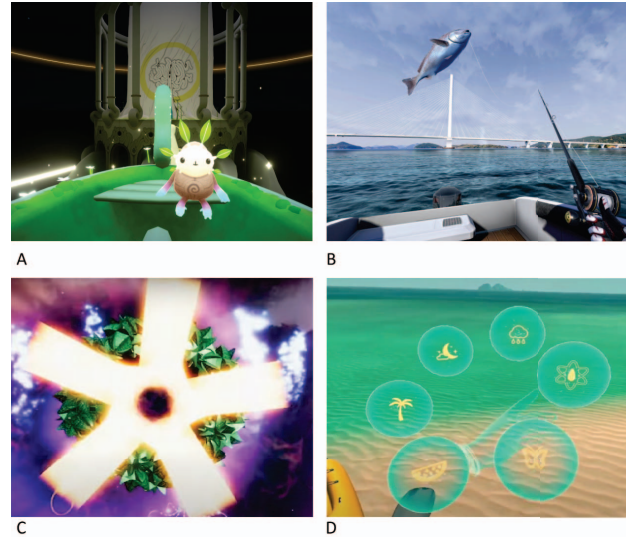


Figure 1: Example screens of the investigated VR experiences: A: Fujii [2], B: Real VR Fishing [9], C: TRIPP [14], D: Nature Treks VR [8].

would win the game?", due to the specifics of the tested games, which could not be won or lost. Finally it consists of 25 items with 5-points Likert scale. To investigate the the current level of anxiety as an emotional state of respondents, a Polish version [8] of the first part of **State-Trait Anxiety Inventory (STAI-1)** [9] was used. It consists of 20 questions with 4-points scale. The **Personal Report of Confidence as a Speaker (PRCS)** is a commonly used measure to assess people's fear of public speaking. We used its shortened version, composed of 12 statements with true-false answers [2]. The semi-structured interview was used as a method to deepen the the observations and opinions. It covered: the justification of the most and least preferred experience, an experience to choose one day / one hour before a stressful event to relax; elements that were most important in the context of relaxation, the biggest distractions that prevented from relaxing, how the best / worst relaxing experience should be like, preferred graphics style.

3 EXPERIMENTAL DESIGN

Before the study, participants filled in an informed consent form and STAI questionnaire. Then they were tasked with testing four games (15 min each) in Meta Quest 2 in a randomized order drawn by them using the Balanced Latin Square. The think-aloud protocol was used. After each test, the participants filled in the immersion and STAI questionnaires. After all experiences, an interview was conducted. The entire study lasted an average of 90 - 120 min. Twelve participants (8 female, 4 male, age 18-36) were recruited

using an online screener form which contained basic demographic information as well as PRCS questionnaire. In order to balance the PRCS among participants, participants with the following rates were selected: 0-3 (3 participants), 3-6 (3), 6-9 (8), 9-12 (5), 12-15 (1), which was close to the Gaussian distribution. Half of the participants already used VR, while for 6 it was their first VR experience.

4 RESULTS AND CONCLUSIONS

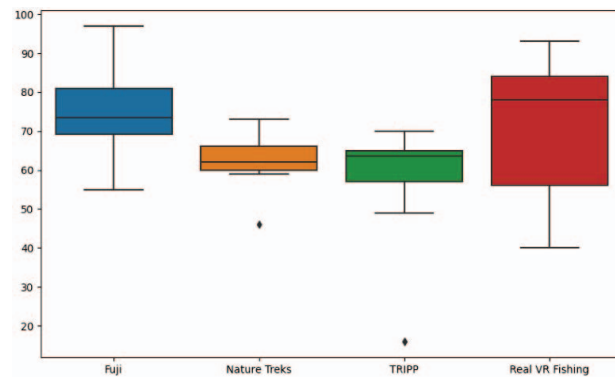


Figure 2: Immersion levels measured for the investigated VR experiences.

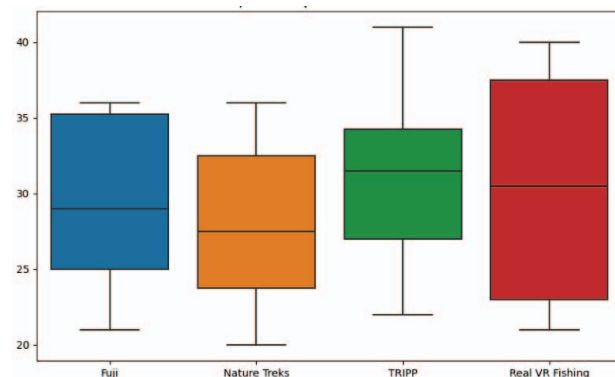


Figure 3: Anxiety levels measured for the investigated VR experiences.

Because of the relatively small sample size, quantitative outcome are interpreted only in the context of qualitative results. Fujii and Real VR Fishing demonstrate higher immersion values, suggesting a more engaging VR experience compared to Nature Treks and TRIPP (Fig. 2). Pre-test results reveal diverse initial anxiety levels - average 34.8 (standard deviation 7.2). Nature Treks and TRIPP appear to lower perceived anxiety, particularly in individuals with initially high anxiety. Fujii and Real VR Fishing elicited varied anxiety responses (Fig. 3).

The data suggests no definitive relation between immersion levels and anxiety. Fujii, with the highest immersion rating, doesn't have the lowest average anxiety level, suggesting that immersion does not necessarily result in less anxiety. The study demonstrated that a game like Nature Treks, despite not being the most immersive (third in the ranking), can induce the lowest anxiety levels among tested games. Opinions on Real VR Fishing varied greatly, with some respondents praising its movement requirements, clear objectives, and graphics, while others found the competitive element stressful. Fujii was often chosen as a favorite for its pleasant music, ambient

sounds, environment interactions, and light tasks that didn't penalize mistakes. TRIPP was frequently criticized for its limited interaction, monotonous tasks, intense music, and lack of customization. Fujii was often chosen as the game to use the day before a stressful event, and Nature Treks as the game to use an hour prior. Notably, Nature Treks was chosen for its ability to provide a short and undemanding disconnection from reality without becoming overly engaging.

As the outcome of the reported investigation, we formulated general recommendations for a relaxing VR experience: **Style:** Some favouring realistic graphics, while others prefer unrealistic ones. As suggested by several respondents, it might be optimal to amalgamate both conventions. **Music:** It's essential for players to be able to select their own type of music, as each individual may find different music types relaxing. There should also be an option to turn off the music. **Gamification:** Tasks, mini-games, or quests in a relaxing game should be voluntary, not imposed by the gameplay. Players may choose to participate in them, but they shall not be obligated to do so. **Simple interactions with the environment:** The ability to immerse oneself in the virtual world and interact with it was crucial for relaxation. A relaxing game should allow for many minor interactions, such as petting animals or grabbing objects.

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